

TRAINING WORKSHOP · 30 MAY 2026

# Workshop on Essentials for Mortality Investigation of Asian Elephant

for Officials of Chhattisgarh Forest Department

**ORGANISED BY**  
**Chhattisgarh Forest Department**  
Bilaspur Circle

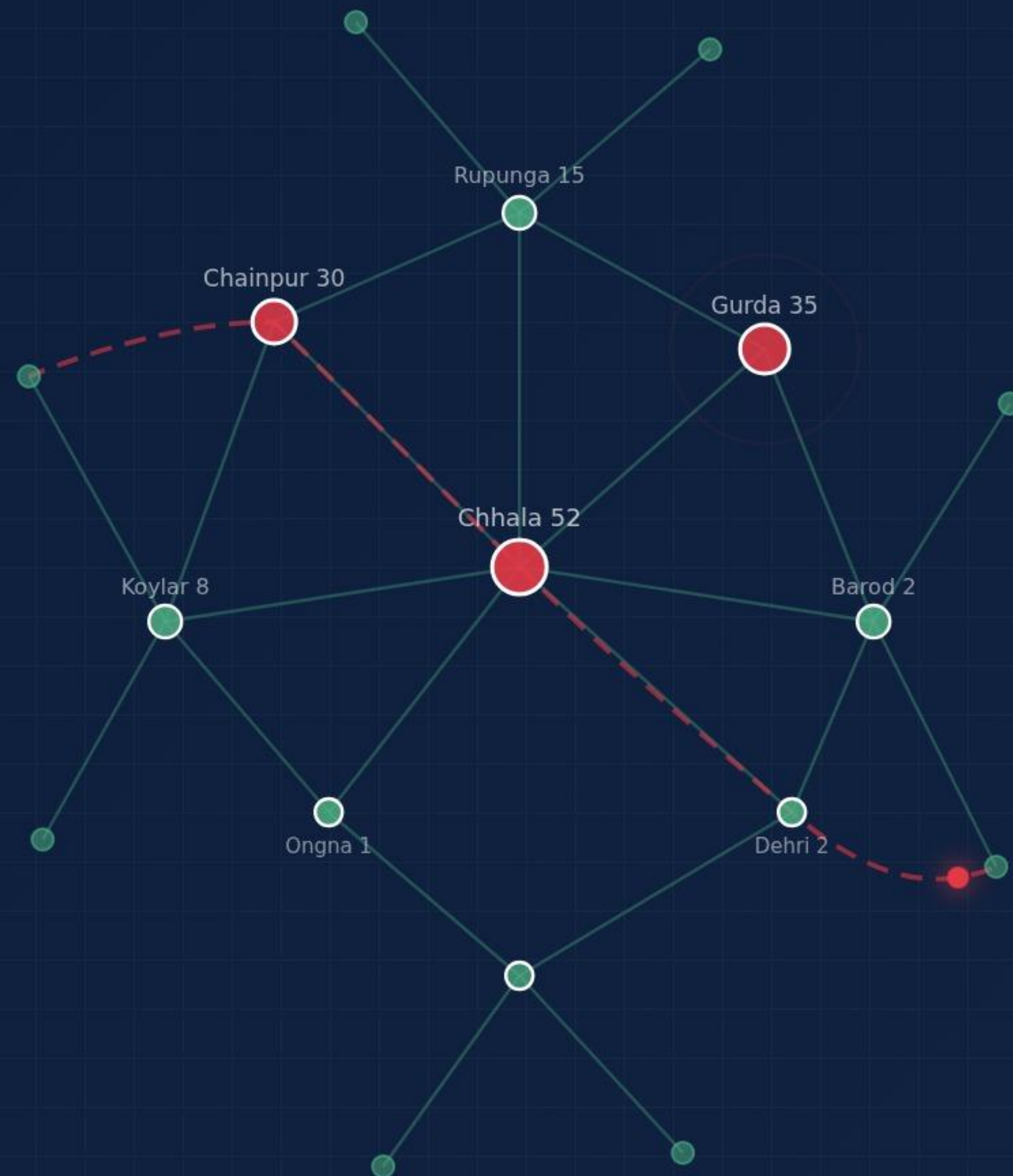
**TECHNICAL PARTNERS**  
**Wildlife Institute of India**  
Dehradun, Uttarakhand  
**Indian Veterinary Research Institute**  
Bareilly, Uttar Pradesh

**Dharamjaigarh** **Raigarh** **2021-2027**



## ELEPHANT MOVEMENT NETWORK

Dharamjaigarh · Raigarh · Bilaspur





# Human-Elephant Conflict: Data, Science & Solutions

3  
OBJECTIVES

3  
OUTCOMES

## — OBJECTIVES OF WORKSHOP

### BIOLOGY, BEHAVIOUR & DISEASE

Build a comprehensive understanding of Chhattisgarh elephant population dynamics, anatomical orientation, and behavioural indicators — covering infectious diseases (**EEHV, tuberculosis, anthrax**) and non-infectious mortality causes including electrocution, train collisions, malicious poisoning, toxicosis, and disaster-induced events.

### NECROPSY, BIOSAFETY & SAMPLE HANDLING

Develop hands-on competency in field necropsy under **biosafety and biosecurity protocols**, vetro-legal investigation with chain of custody documentation, and rigorous sample collection, fixation, cold chain management, and transport logistics to reference laboratories.

### SURVEILLANCE, DISPOSAL & REMEDIATION

Establish operational frameworks for **active and passive health surveillance**, implementing standardized protocols for safe carcass disposal, chemical site decontamination, and environmental remediation to protect local ecosystems.

## — EXPECTED OUTCOMES

### FIELD NECROPSY SKILLS

Field veterinarians will be proficient in practical skills needed to properly **position and systematically eviscerate heavy elephant carcasses** in remote settings.

### SAMPLE & LAB PROTOCOLS

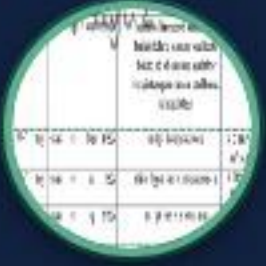
Hands-on experience with **tissue fixative formulation and secure packaging** to ensure diagnostic samples reach reference laboratories without degradation or cross-contamination.

### SURVEILLANCE & FORENSICS

Implementation of **field-validated carcass disposal and site decontamination** protocols, standardized forensic reporting, and proactive health monitoring to rapidly detect emerging threats within the state's elephant populations.



## RESOURCE PERSONS



**Dr. Parag Nigam**

Scientist G & Head, Dept. of Wildlife Health Management  
ICAR-IVRI, Izatnagar

**SESSION CHAIR**

**TS**

**Dr. Tapendra Saini**

Scientist-C  
Dept. of Wildlife Health Management

**RESOURCE PERSON**

**CS**

**Dr. C.P. Sharma**

Principal Technical Officer  
Forensic Cell

**RESOURCE PERSON**

**MK**

**Dr. M. Karikalan**

Senior Scientist  
Centre for Wildlife

**RESOURCE PERSON**

**AS**

**Dr. A.B. Shrivastav**

Former Director  
Centre for Wildlife & Forensic Health

**RESOURCE PERSON**



# Two-Day Workshop Programme

| DAY 1 30 May 2026 |                                                                                                                                                                                          | 10:00 – 20:00+ |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| 10:00<br>11:00    | <b>Inauguration</b><br>Official opening, workshop objectives, and introduction of experts & participants.                                                                                |                |
| ~11:00            | ☞ <b>High Tea &amp; Group Photograph</b>                                                                                                                                                 |                |
| 11:00<br>11:30    | <b>Elephant Status &amp; Human-Elephant Interface in Chhattisgarh</b><br>Population dynamics, corridor use, conflict and mortality trends.<br><br><span>PP, AM</span>                    |                |
| 11:30<br>12:15    | <b>Biological &amp; Anatomical Aspects of Elephants</b><br>Key anatomical peculiarities relevant to field investigation.<br><br><span>PN</span>                                          |                |
| 12:15<br>13:00    | <b>Understanding Wildlife Mortalities: The Veterinary Perspective</b><br>Carcass condition, natural vs unnatural death; ante/post-mortem changes & time of death.<br><br><span>PN</span> |                |
| 13:00<br>14:00    | 🍽 <b>Lunch</b>                                                                                                                                                                           |                |
| 14:00<br>15:30    | <b>Infectious Causes of Mortality in Elephants</b><br>Fatal infectious diseases: EEHV, EMC, FMD, Rabies, TB, Anthrax, Haemorrhagic Septicaemia.<br><br><span>KM, ABS</span>              |                |
| 15:30<br>16:30    | <b>Non-Infectious &amp; Disaster Mortalities in Elephants</b><br>Electrocution, lightning, train hits, infighting, and other natural disaster deaths.<br><br><span>ABS, KM</span>        |                |
| 16:00<br>16:15    | ☞ <b>Tea Break</b>                                                                                                                                                                       |                |
| 16:30<br>17:30    | <b>Non-Infectious Diseases in Elephants</b><br>Mycotoxicosis, kodo millet poisoning, and environmental bio-toxins.<br><br><span>KM, PN</span>                                            |                |
| 20:00+            | 🍽 <b>Course Dinner</b>                                                                                                                                                                   |                |

| DAY 2 31 May 2026 |                                                                                                                                                                                                              | 09:30 – 17:30 |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| 09:30<br>10:15    | <b>Biosafety &amp; Biosecurity Measures in Elephant Mortality</b><br>Disease threats from carcasses, essential PPE, safe disposal, site sanitation & disinfection.<br><br><span>KM, TS, PN</span>            |               |
| 10:15<br>11:00    | <b>Equipment Familiarization &amp; Site Preparation</b><br>Heavy-duty necropsy equipment (chain saws, axes, winches) and establishing a secure perimeter.<br><br><span>ABS, KM, TS</span>                    |               |
| 11:00<br>11:30    | ☞ <b>Tea Break</b>                                                                                                                                                                                           |               |
| 11:30<br>12:30    | <b>Comprehensive Sampling Protocols</b><br>Collection, packaging, cold chain, transport & documentation. Tissue selection for histopathology, toxicology & microbiology.<br><br><span>KM, ABS, PC, TS</span> |               |
| 12:30<br>13:00    | <b>Recording Information During Necropsy</b><br>Report writing and photographic documentation standards.<br><br><span>KM, ABS</span>                                                                         |               |
| 13:00<br>14:00    | 🍽 <b>Lunch</b>                                                                                                                                                                                               |               |
| 14:00<br>14:45    | <b>Vetro-Legal &amp; Forensic Aspects of Mortality Investigation</b><br>Investigation essentials for legal and forensic frameworks.<br><br><span>CP</span>                                                   |               |
| 14:45<br>16:00    | <b>Practical: Crime Scene Investigation</b><br>Chain of custody, evidence collection, and legal procedures for Ivory/tusk extraction.<br><br><span>CP, PN, AM, TS</span>                                     |               |
| 16:00<br>16:15    | ☞ <b>Tea Break</b>                                                                                                                                                                                           |               |
| 16:15<br>16:45    | <b>Panel Discussion</b><br>Critical aspects of managing mortalities in elephants.<br><br><span>ABS, PN, KM, PP, AM, JK</span>                                                                                |               |
| 16:45<br>17:30    | <b>Valedictory</b><br>Workshop recap, feedback collection, and closing ceremony.                                                                                                                             |               |



## ATTENDING DELEGATIONS

# 48

TOTAL PARTICIPANTS

### Dharamjaigarh Van Mandal

DFO, Range Officers, Beat Guards

### Raigarh Van Mandal

DFO, Range Officers, Beat Guards

### ICAR-IVRI Izatnagar

Senior Scientists, Lab Analysts

### Wildlife Department CG

Wildlife Wardens & Inspectors

### GIS Cell, Bilaspur

GIS Analysts & Cartographers

### District Administration

Revenue & Tehsil Officers

CURRENT STATUS

# Elephant Population Chhattisgarh

ESTIMATED 2025

**451**

Elephants in CG

SINCE 2001

**+13%**

Annual growth (CAGR)

- Population grew from **24 (2001)** to **~451 (2025 est.)** over two decades
- Bilaspur District hosts the largest concentration of elephants in CG
- Seasonal corridors extend into Raigarh, Korba and Surguja districts
- Human-elephant interface intensifying as agricultural land expands into corridors
- 2025-26 records the highest single-year conflict events in the five-year study period

POPULATION TREND 2001-2025

# Elephant Range — Bilaspur District

Approximate range  
boundary across  
Dharamjaigarh &  
Raigarh Van Mandals.

**Dharamjaigarh** — 27 elephant  
deaths · 36 human deaths  
**Raigarh** — 21 elephant deaths ·  
9 human deaths



LIVE ELEPHANT TRACKING · 31 MAY 2026

154

Elephants Currently Tracked

DHARAMJAIGARH

113

12 Locations

RAIGARH

41

4 Locations

NOTABLE LARGE HERDS

|                         |    |
|-------------------------|----|
| Edu, Chhala Range       | 52 |
| Chainpur, Dharamjaigarh | 30 |
| Gurda, Kharsiya (RG)    | 35 |

CURRENT GPS LOCATIONS · 16 SITES

• LIVE

● Current elephant location (icon size = herd count) ● Pulsing = large herd (≥30)

←

1 / 22

→

Edit



ELEPHANT CASUALTIES

0

Total Elephant Deaths

2021-22 to 2026-27 · Both Divisions

DHARAMJAIGARH

27

DH Division

RAIGARH

21

RG Division

RECORD — 2025-26 · RISING — 2026-27

2025-26: 12 deaths (record year) · 2026-27: 4 deaths in first 2 months — all calves, drowning.

YEAR-WISE DISTRIBUTION

48

Total Deaths

Last bar (2026-27\*) is current year — ongoing, figure may rise.



1 / 22



Edit



# Cause of Death — Both Divisions Combined

51%  
Electrocution

# Elephant Casualties — Age & Sex Profiling

Age Group   Sex Group   Seasonal   Range-wise

39% under 15 years (calves + young) — elevated mortality in young age classes indicates a long-term conservation concern for population viability.



# Elephant Casualties — Age & Sex Profiling

Age Group   Sex Group   Seasonal   Range-wise

# Elephant Casualties — Age & Sex Profiling

Age Group   Sex Group   **Seasonal**   Range-wise

**Peak: October (8 deaths)** — post-monsoon movement & harvest season overlap

**Oct-Dec = 42% of annual deaths**



# Elephant Casualties — Age & Sex Profiling

## Casualty Map — Both Divisions

100% Geo-referenced coverage of all incidents 2021-26.

Click any marker for details

● Elephant Death (48)

● Human Death (45)

○ Total: 86 geo-points



# Postmortem Report — Summary Analysis

## Coverage & Compilation

41 elephant mortality records compiled across Dharamjaigarh and Raigarh Van Mandals for the period 2021-2026. All available case records have been cross-referenced with the geo-referenced incident database.

## Lead Cause — PM Confirmed

Electrocution leading to Cardio-Respiratory Failure confirmed as cause of death in 20 of 41 records (49%). Multiple simultaneous electrocution events noted in 3 cases. Drowning is the second-leading confirmed cause (9 of 41, 22%).

## Age Concern — Neonates & Calves

Neonates and calves (under 1 year) account for approximately 26% of known-age records (10 of 39). Ten neonate and calf cases identified. This level of juvenile mortality poses a direct threat to long-term population viability.

## PM Cause Distribution — 41 Records

| CAUSE             | PM CONFIRMED | % OF 41 | STATUS      |
|-------------------|--------------|---------|-------------|
| Electrocution     | 20           | 49%     | PREVENTABLE |
| Drowning          | 9            | 22%     | PREVENTABLE |
| Natural / Old Age | 5            | 12%     | NATURAL     |
| Trauma / Fall     | 5            | 12%     | INVESTIGATE |
| Other             | 2            | 5%      | REVIEW      |



KEY PM FINDINGS

# Five-Year Trend & PM Highlights

- **Sex distribution — near equal:** 20 male / 18 female from 38 sexed records (3 unknown). No strong sex-biased conflict mortality.
- **Multiple neonates & calves** — 10 calf/neonate cases confirmed. Under-1-yr mortality at 26% of known-age records (10 of 39) is alarming.
- **3 simultaneous electrocution events** — single incidents killed multiple elephants; indicates systemic power-line hazard.
- **Vehicular collision confirmed** — 1 case (2023-24). Infrastructure corridor risk identified and documented.
- **Inter-elephant conflict** — 1 confirmed case in 2023-24, Dharamjaigarh, possibly musth-related.
- **2025-26 recorded 12 deaths** — highest year on record. 2026-27 has already recorded 4 deaths in first 2 months, all calves drowned.

YEAR-ON-YEAR TREND

Both elephant and human casualties show a **rising trend** with 2025-26 being the worst year on record for both categories.

YEAR-WISE CASUALTY TREND



## LAB COVERAGE

## 6 Lab Reports — 12 Animals Covered

Six batch reports from ICAR-IVRI Izatnagar cover 12 individual elephant deaths across the study period. Reports compiled under File F.2-19/DI/NRC/2025-26/CWL. Dr. M. Karikalan is the signing authority for all reports.

## ARSENIC DETECTED

## Arsenic in 2 Specimens — December 2025, Tamnar Range

Arsenic detected at elevated levels in visceral samples from 2 specimens from Tamnar Range (Raigarh Division), December 2025. Hepatocyte degeneration confirmed on histopathology. Indicative of contaminated water source or feed near industrial corridor. Environmental sampling of local water bodies recommended immediately.

## CRITICAL GAP

## Missing Report — ID 5008/2025

The laboratory report for specimen ID 5008/2025 is MISSING from the consolidated file. Both adjacent specimens (same batch, same location) showed arsenic positivity. This gap must be closed — the originating file should be obtained directly from ICAR-IVRI Izatnagar for completeness and medico-legal purposes.

## TOXICOLOGY — NEGATIVE

## Broad Toxicological Screen — 5 Batches

Five tested batches returned **NEGATIVE** results for: Nitrate-Nitrite compounds, Heavy metals (Pb, Hg, Cd), Organochlorine & organophosphate insecticides, Hydrogen Cyanide (HCN). This clears the most common pesticide/industrial toxin hypotheses.

## EEHV — NEGATIVE

## Elephant Endotheliotropic Herpesvirus (EEHV)

EEHV PCR testing returned **NEGATIVE** in all tested samples. No viral hemorrhagic disease involvement confirmed. This is particularly significant given the high neonate mortality — viral aetiology is ruled out for current cases.



**2025-26 is the worst year  
on record.**

**Electrocution is  
preventable.**

**Act now.**



### **Fix Power Lines**

Audit & upgrade all low-hung and illegal electric fencing within 5 km of elephant corridors. 51% of deaths are preventable.



### **Install Escape Ramps**

Ramps at Panikshet Dam, Rabo Dam and all large water bodies in conflict zones. Drowning = 22% of deaths.



### **Deploy Early Warning**

SMS-based real-time elephant movement alerts for villages in Chhal and Ghargoda beats. Protect human lives



HUMAN CASUALTIES

0

Human Lives Lost to Elephant Encounters

2021-22 to 2026-27 · Both Divisions

DHARAMJAIGARH

36

DH Division

RAIGARH

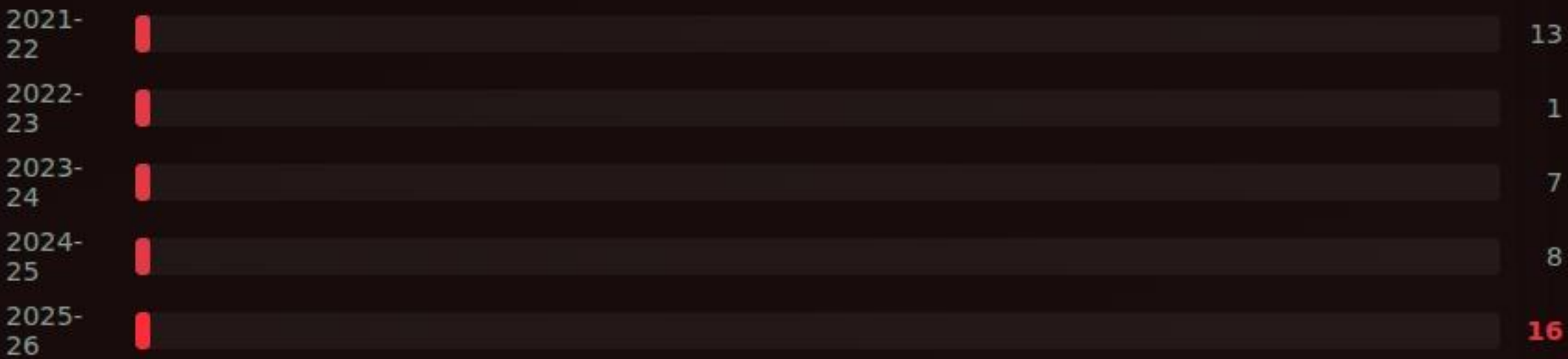
9

RG Division

HOTSPOT BEATS

Chhal Beat: 10 · Borojh: 9 · Lailungaan: 8

YEAR-WISE HUMAN DEATHS



2025-26: 16 deaths — the highest single-year human toll in the study period. DH Division alone accounts for 80% of all human casualties.

Most incidents occur during night hours near agricultural fields, especially during harvest season (October-December), when elephant movement peaks.

# Combined Casualty Map

90 Geo-referenced Points — Both Divisions, 2021–2026

Elephant markers appear first.  
Human markers appear after 2 seconds.  
Hotspot pulses mark critical zones.

- Elephant Death (48)
- Human Death (45)
- Critical Hotspots



END OF PRESENTATION

# Thank You

Together for Coexistence

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## Forest Department, Bilaspur

Dharamjaigarh Van Mandal · Raigarh Van Mandal

## Data Sources

Official Proforma-1 Records, Dharamjaigarh & Raigarh Divisions  
PM Reports: ICAR-IVRI Izatnagar (File F.2-19/DI/NRC/2025-26/CWL)  
GIS Data: Bilaspur Forest Division, GIS Cell

**Generated:** 30 May 2026

**Dharamjaigarh**  
Van Mandal

**Raigarh**  
Van Mandal

**ICAR-IVRI**  
Izatnagar

**GIS Cell**  
Bilaspur